

GOPIKA P S

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PROFESSIONAL SUMMARY

Results-driven Electronics and Communication Engineering student with hands-on experience in Embedded Systems, FPGA development, and IoT architectures. Proficient in leveraging microcontrollers (STM32, Arduino) and hardware description languages (Verilog) to design real-time monitoring and automation solutions. Adept at circuit prototyping and PCB assembly, seeking to apply analytical problem-solving skills to complex engineering challenges.

EDUCATION

B.Tech, Electronics & Communication Engineering

Expected 2027

Government Engineering College, Sreekrishnapuram, Palakkad
APJ Abdul Kalam Technological University, Kerala
CGPA : 9.19 (up to semester 5)

Higher Secondary Education - Biology Science

2020 - 2022

St. John's Higher Secondary School, Undancode
Percentage : 95.16%

TECHNICAL SKILLS

Hardware & Core Concepts:

Embedded Systems, IoT, Digital Electronics, FPGA
Architecture, PCB Design, Sensor Interfacing, Circuit
Prototyping, Finite State Machines (FSM), Testbench Design
C/C++, Python (Scientific Computing & ML), HTML, Verilog
Xilinx Vivado, STM32CubeIDE, Proteus, Icarus Verilog,
Arduino IDE, Tinkercad
Microsoft Office Suite, WordPress, Thunkable, GTKWave,
LTSpice

Programming Languages:

Engineering Software:

Tools & General:

TECHNICAL PROJECTS

Adaptive Real-Time Traffic Light Control Using FPGA | Zedboard • FSM • Verilog • Vivado

- Engineered an intelligent adaptive traffic control system, as measured by functional real-time signal management, by utilizing a ZedBoard with Zynq-7000 SoC.
- Designed complex Finite State Machines (FSM), as measured by zero-defect verified signal integrity, by utilizing Verilog HDL and simulating hardware behaviors with GTKWave.

Temperature & Humidity Monitoring System | Arduino • Sensor Integration

- Programmed a responsive environment monitoring tool, as measured by continuous and reliable recording of climatic data, by integrating hardware sensors with an Arduino development board.

EnviroGuard: IoT Air Quality Monitor | STM32 • IoT • Proteus

- Developed an IoT-based air quality monitor, as measured by accurate real-time tracking of environmental metrics, by integrating STM32 microcontrollers with precision environmental sensors.
- Optimized prototyping budget constraints, as measured by an early and cost-efficient hardware deployment, by conducting comprehensive component cost analysis and structural planning.

INTERNSHIP EXPERIENCE

Analog IC Design Trainee Intern

June 2026

Sophrosyne Semiconductor, Kunjibettu, Udupi

- Studied the complete Analog IC Design Flow, including schematic design, layout, physical verification, and post-layout validation.

- Learned PDKs, technology libraries, device libraries, simulation models, DRC, LVS, ERC, and PEX methodologies used in semiconductor design.
- Explored analog building blocks such as MOSFETs, current mirrors, differential amplifiers, resistors, capacitors, and diodes.
- Gained understanding of device matching, common-centroid layout, guard rings, corner analysis, Monte Carlo analysis, and reliability concepts

PCB Design & Assembly Intern

June 2025

Cyborg AI Gen Automation, Kochi, Kerala

- Executed precise custom PCB design, as measured by the successful fabrication of functional electronic prototypes, by translating schematic designs into physical hardware.
- Assembled complex circuitry, as measured by the optimal integration of embedded components, by applying industry-standard PCB manufacturing techniques.

WORKSHOPS & CERTIFICATIONS

Embedded System Communication Protocols – Hands-on training at TBI GEC Sreekrishnapuram (2026)

Artificial Intelligence with Python – Techmaghi & KSHITIJ IIT Kharagpur (2024)

Cyber Security & Ethical Hacking – Techmaghi & Tryst'24 IIT Delhi (2023)

Interfacing with the Raspberry Pi – University of California, Irvine (Coursera, 2025)

Arduino Platform and C Programming – University of California, Irvine (Coursera, 2025)

Introduction to IoT and Embedded Systems – University of California, Irvine (Coursera, 2025)

Interfacing with the Arduino – University of California, Irvine (Coursera, 2025)

Problem Solving Innovation – Wadhwani Foundation (2025)

ACHIEVEMENTS & LEADERSHIPS

- Achieved the prestigious Scouts and Guides Rajyapuraskar Award and secured the National Cadet Corps (NCC) 'B' Certificate, as measured by consistent leadership evaluations, by completing rigorous training programs.
- Facilitated technical community engagement, as measured by successful regional program execution, by operating as a Volunteer for the IEEE Young Buddies 2.0 Program.
- Collaborated across multiple engineering societies (Robotics Club, IEEE, ISTE, Mu-learn), as measured by active knowledge sharing initiatives, by participating in regular technical problem-solving sessions.

LANGUAGES

Malayalam : Native **English** : Working Proficiency **Tamil** : Basic Proficiency

TECHNICAL INTERESTS

Core Areas : VLSI • Digital Design • Embedded Systems • FPGA Designs • IoT • PCB Designing